Published Preview Edit

Exam 1 - Part C

This is a preview of the draft version of the quiz

topics with anyone other than course staff. While you are permitted to look things up on the web, you may not post questions related to the exam (except to Course Staff - send private messages on Piazza).

Before beginning this Quiz, check to see if any announcements have been made on Canvas. (you don't want to do this once the clock starts ticking when you begin the Quiz).

If you have a technical problem, try to resolve the issue yourself and proceed with the quiz as much as possible. Your 20 minutes do not pause once they start. As soon as possible, send a message to course staff. If you send a message (privately on Piazza), please provide as much detail as you can about your problem, and include your email address.

Quiz Type	Graded Quiz
Points	31
Assignment Group	Exams
Shuffle Answers	No
Time Limit	20 Minutes
Multiple Attempts	No
View Responses	Always
Show Correct Answers	Immediately
One Question at a Time	No

Due	For	Available from	Until
Mar 8 at 9pm	Everyone	Mar 8 at 5:30pm	Mar 8 at 9pm

Preview

Score for this quiz: **0** out of 31

Submitted Mar 8 at 1:45pm

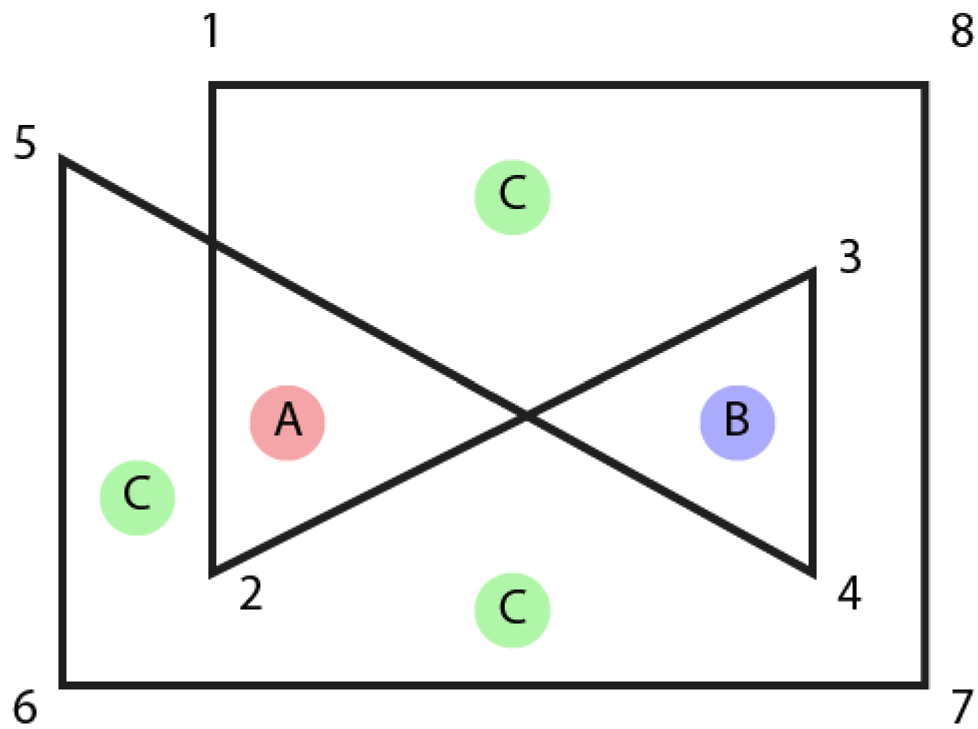
This attempt took less than 1 minute.

Jnanswered

Question 1

0 / 4 pts

The following polygon (it has 8 vertices) is created in Canvas. The black line is drawn
when context.stroke() is called. When context.fill() is called to fill this polygon, the points A, B and
C may or may not be filled, depending on the filling rule. Note: the 3 Cs are all in the same “region”
so they will all have the same answer.
Check all that apply.



orrect Answer

- [ ] A will be filled with the "nonzero" rule (the default rule)
- [ ] A will be filled with the "evenodd" rule
- [ ] B will be filled with the "nonzero" rule (the default rule)
- [ ] B will be filled with the "evenodd" rule

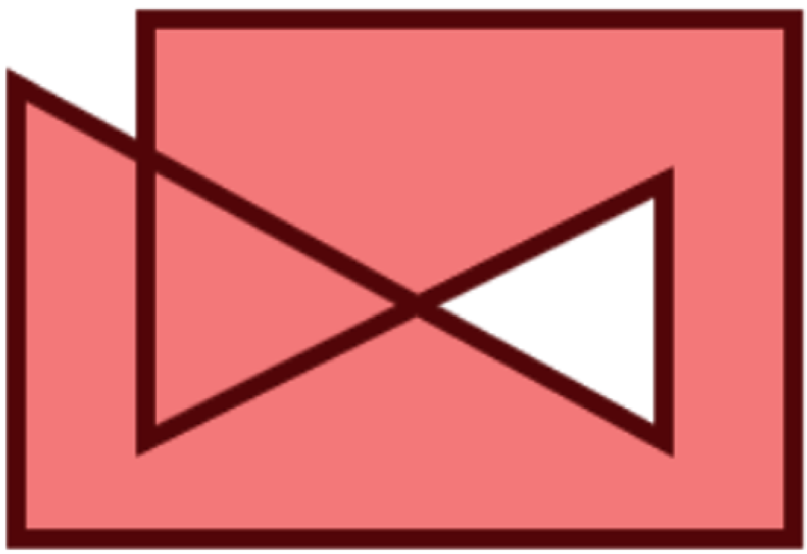
orrect Answer

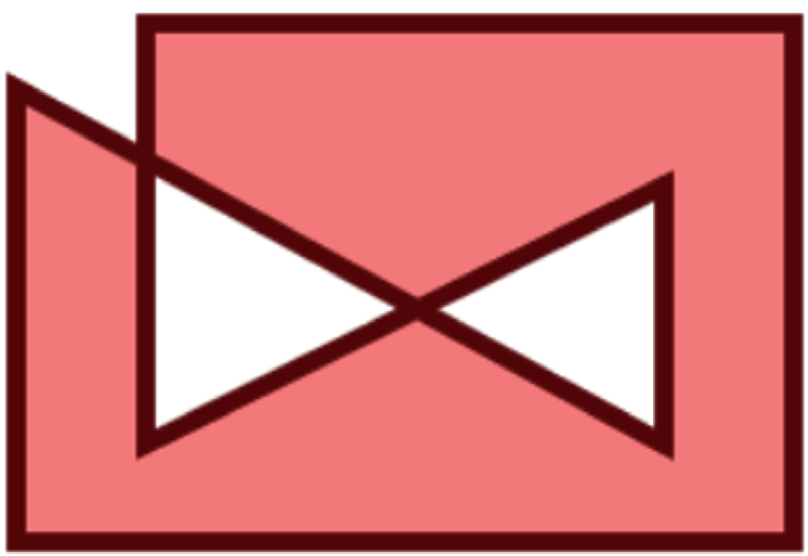
- [ ] C will be filled with the "nonzero" rule (the default rule)

orrect Answer

- [ ] C will be filled with the "evenodd" rule

Even Odd should be easy - start at A or B, you cross 2 lines in most directions - so its inside.
A and B for non-zero are tricky.
Each point will be circled twice. If you like, think about it as an outer polygon (1,X,5,6,7,8) and 2
triangles (X,2,Y) and (3,4,Y) where X and Y are "intersection points".
The outer polygon is counter-clockwise.
The triangle around A is counter clockwise.
The triangle around B is clockwise.
A has 2 loops in the same direction - winding is 2, it is non-zero.
B has 2 loops in different directions (+1 and -1) - it is zero.





The following three questions refer to this javascript program:

```
console.log("Start");

function one() {
    console.log("One");
}
function two() {
    console.log("Two");
}
function three() {
    console.log("Three");
}

window.requestAnimationFrame(one);
two();
console.log("End");
```

Jnanswered

Question 2

0 / 2 pts

The word "One" is

- Always printed before the word "End"
- Sometimes printed before "End", but sometimes afterwards

orrect Answer

- Always printed after the word "End"
- Printed more than once by this program
- Never printed by the program

Jnanswered

Question 3

0 / 2 pts

The word "Two" is

- Printed after the word "One"
- orrect Answer Printed before the word "End"
- Printed before the word "Start"
- Printed more than once by this program
- Never printed by this program

Jnanswered

Question 4

0 / 2 pts

The word "Three" is

- Printed before "Start"
- Printed after "Start" but before "End"
- Printed after "End"
- orrect Answer Never printed by this program

Jnanswered

Question 5

0 / 4 pts

The code segment:

```
context.rotate( Math.PI /2 ); // 90 degrees
context.translate( 10, 0 );
context.rotate( -Math.PI /2 ); // -90 degrees
context.translate( -10, 0);
```

is equivalent to:

```
context.translate( [ ] , [ ] );
```

Answer 1:

ou Answered

(You left this blank)

orrect Answer

-10

Answer 2:

ou Answered

(You left this blank)

orrect Answer

10

Jnanswered

Question 6

0 / 3 pts

This program draws a triangle (it really does - I tried it).

```
context.strokeStyle = "black";
context.save();
context.beginPath();
context.moveTo(20,20);
context.strokeStyle = "red";
context.lineTo(100,20);
context.strokeStyle = "green";
context.stroke();
context.lineTo(60,80);
context.strokeStyle = "blue";
context.stroke();
context.restore();
context.beginPath();
context.moveTo(60,80);
context.lineTo(20,20);
context.stroke();
context.strokeStyle = "purple";
```

Which of the colors appear in the final result? (check all that apply)

orrect Answer

- [ ] black
- [ ] red
- [ ] green

orrect Answer

- [ ] blue
- [ ] purple

The program does draw green, but it is covered over by blue.

The restore returns things to black, and the last segment is drawn before switching the pen to purple.

Jnanswered

Question 7

0 / 4 pts

Two curves meet such that the end of 1 curve is the beginning of the next.

You may assume that no place on either curve does the 1st, 2nd, or 3rd derivative vanish. (note: this fact is only important if you care about an obscure detail of the definitions of G continuity that is described in the book but doesn't mean much in practice)

Which of the following is possible?

orrect Answer

- [ ] The curves are G(2) but not C(2)
- [ ] The curves are C(2) but not G(2)
- [ ] The curves are C(2) but not C(1)
- [ ] The curves are G(2) but not G(1)

orrect Answer

- [ ] The curves are G(2) but not C(1)

In class, we said that almost always C(n) implies G(n) (since if the derivatives are the same, then they must be going in the same direction).

There is a weird case that if the derivative is zero, it doesn't have a direction - but that's the weird case of the "except if the parametric derivative vanishes".

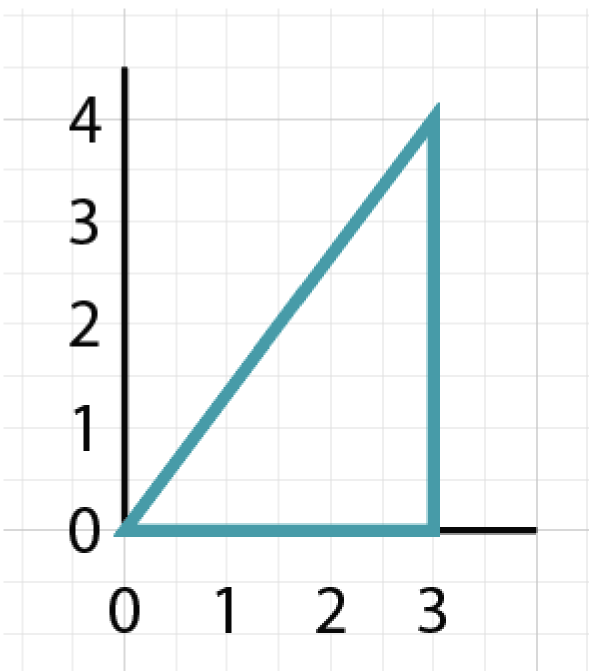
Also, while C(n) implies C(n-1) (or G(n) implies G(n-1)) the reverse is not true.

Jnanswered

Question 8

0 / 4 pts

A curve goes around the triangle (0,0), (3,0), (3,4) and back to (0,0) in that order. It is arc-length parameterized (so the curve makes a full loop starting from u=0 to u=1, at both u=0 and u=1, the curve is at (0,0).



At u=.5, the curve is at ____ , ____ ?

Answer 1:

ou Answered

(You left this blank)

orrect Answer

3

Answer 2:

ou Answered

(You left this blank)

orrect Answer

3

This is a 3,4,5 triangle, so the total length is 12.

Half way along the length is 6 units, which is 3/4 of the way up the vertical edge.

Jnanswered

Question 9

0 / 6 pts

The following code snippet:

```
context.translate(3,5);
context.scale(2,2);
context.translate(3,5);
context.scale(1,2);
```

could be written as the following (fill in the blanks):

context.translate(____ , ____);

context.scale(____ , ____);

Answer 1:

ou Answered

(You left this blank)

orrect Answer

9

Answer 2:

ou Answered

(You left this blank)

orrect Answer

15

Answer 3:

ou Answered

(You left this blank)

orrect Answer

2

Answer 4:

ou Answered

(You left this blank)

orrect Answer

4

Quiz Score: **0** out of 31